UCLA Extension

Introduction to Data Science COM SCI X 450.1

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HOMEWORK 2

NOTE: for this homework assignment, please use the `airquality, CO2` and `mtcars` base R data sets where appropriate.

IMPORTANT: Please use Canvas to submit a single R script file with your answers. Include both the **R code and output** (as comments) **for each question**. There is no need to submit image files containing the plots.

IMPORTANT: Please do NOT use a large language model like ChatGPT to do your homework.

IMPORTANT: Please work on the homework independently. This is NOT a group assignment.

IMPORTANT: Please submit your original work ONLY.

Question 1

Using the `sqldf()` function found in the `sqldf` package to select data from the CO2 data set, execute the SQL statement required to calculate the average value for `uptake` grouped by `Type`. Please use only SQL for the solution to this question. [Hint: use Google to find tutorials for using SQL with the `sqldf` package.]

Question 2

Use the following vector assignment statements to provide data content for a new data frame:

```
Died.At <- c(22,40,72,41)
Writer.At <- c(16, 18, 36, 36)
First.Name <- c("John", "Edgar", "Walt", "Jane")
Second.Name <- c("Doe", "Poe", "Whitman", "Austen")
Sex <- c("MALE", "MALE", "MALE", "FEMALE")
Date.Of.Death <- c("2015-05-10", "1849-10-07", "1892-
03-26","1817-07-18")
```

Write some data munging code to performing the following operations:

- Create a new data frame `df` with the above data for each of six columns.
- Use the appropriate `as.()` function to cast the `Sex` variable to a factor.
- The variable names are inconvenient so write R code to change them to: `age_at_death`, `age_as_writer`, `first_name`, `surname`, `gender`, `date_died` [Hint: remember the `names()` function for data frames.]
- Say "John Doe" died on his birthday, calculate and display his birthdate value (only John's) based on the variables `date_died` and `age_at_death`

Question 3

When recording experimental observations, there are two general formats found in data sets – "long" and "wide." The long format for recording observations is when there is one observation row per variable. A lot of statistical tests favor this format. Here is an example of wide format:

product	height	width	weight
A	10	5	2
B	20	10	NA

The following R code generates a data frame in wide format with some simulated data:

```
> product <- c("A", "B")

> height <- c(10,20)

> width <- c(5,10)

> weight <- c(2,NA)

> observations_wide <- data.frame(product, height,
width, weight)

> observations_wide

  product height width weight
1       A     10     5      2
2       B     20    10     NA
```

The wide format for recording observations is when you have multiple values, spread out over multiple columns, for the same observations. Since different functions may require you to input your data either in long or wide format, you might need to *reshape* your data set.

Write a data transformation R script to take the `observations_wide` data frame above and convert it to long format. Here is what the output should look like below (make sure you order the rows to match the results shown).

product	variable	value
A	height	10
A	width	5
A	weight	2
B	height	20
B	width	10

[Hint: take a look at `reshape2` package and the `melt()` function, which should also handle the removal of NAs]

Question 4

Using the `mtcars` data set, write an R script that calculates the average miles per gallon (`mpg` variable) by number of cylinders in the car (`cyl` variable). The output should be the following:

```
        4         6         8
26.66364 19.74286 15.10000
```

[HINT: one solution uses a loop function along with `split()`]

Question 5

Using the `mtcars` data set, write an R script to calculate the absolute difference between the average horsepower of 4-cylinder cars and the average horsepower of 8-cylinder cars. [Hint: you may wish to use the base R `abs()` function for calculating the absolute value of a number.]

Question 6

Using the `airquality` data set, provide the R code that calculates mean value of the `Temp` variable when the `Month` variable is equal to 6?

Question 7

Write the `dplyr` code required to calculate the mean `mpg` for each transmission type (0 = automatic, 1 = manual) using the `am` variable in the `mtcars` data set. Sort the resulting list by mean `mpg`. Use a single `dplyr` statement with multiple pipes in the solution. The resulting output should be:

```
# A tibble: 2 x 2
#    am mean_mpg
# <dbl>    <dbl>
#1    0     17.1
#2    1     24.4
```

[Hint: considering using the following `dplyr` functions: `group_by`, `summarize`, and `arrange`].

Question 8

In this question you'll need to use the `scatterplot3d` package in order to render a 3D visualization for the `mtcars` data set. You'll need to use the `scatterplot3d()` function in this package. Please provide the R code that produces a data visualization with the following requirements:

- Use the variables for the X, Y and Z axis respectively: `wt, disp, mpg`
- Include an appropriate title for the plot
- Include labels for each axis and include the units for the variables
- Add a fourth data point to the plot, i.e. the `am` variable (transmission), by using the `pch` argument.
- Please specify each argument name when calling the function.

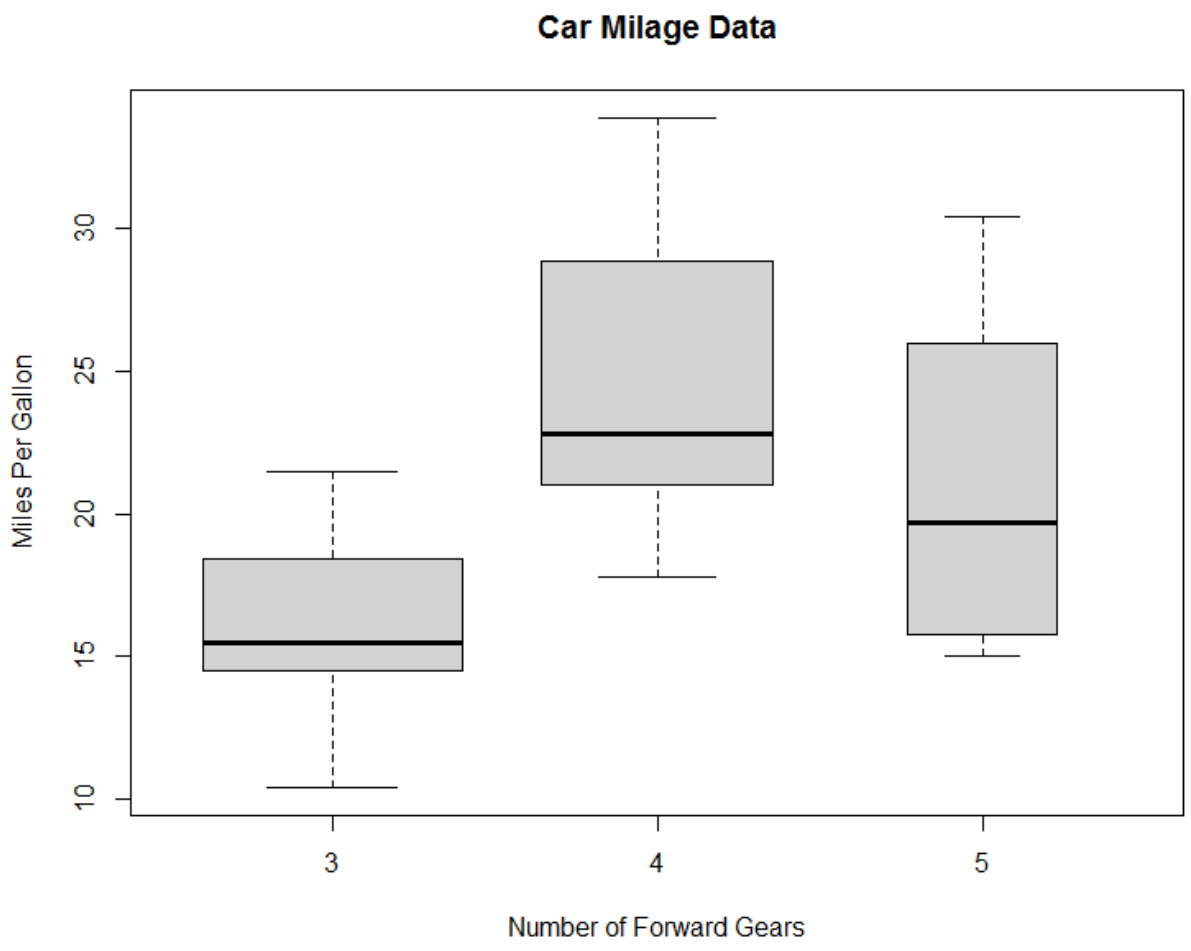
Question 9

Using the `CO2` data set, produce a histogram for the `uptake` variable with the following specifications:

- Use 20 cells (aka buckets, breaks) for the plot
- Use the color name "cornflowerblue" for the plot
- Store the histogram in an object variable
- Extract from the object and display the break points (cell boundaries)
- Extract from the object and display the counts for each cell

Question 10

Provide the R code necessary to reproduce the boxplot data visualization below using the `mtcars` data set. [HINT: make sure the width of the boxes represents the number of observations in the groups]


Car Milage Data
Miles Per Gallon
10
15
20
25
30
3
4
5
Number of Forward Gears